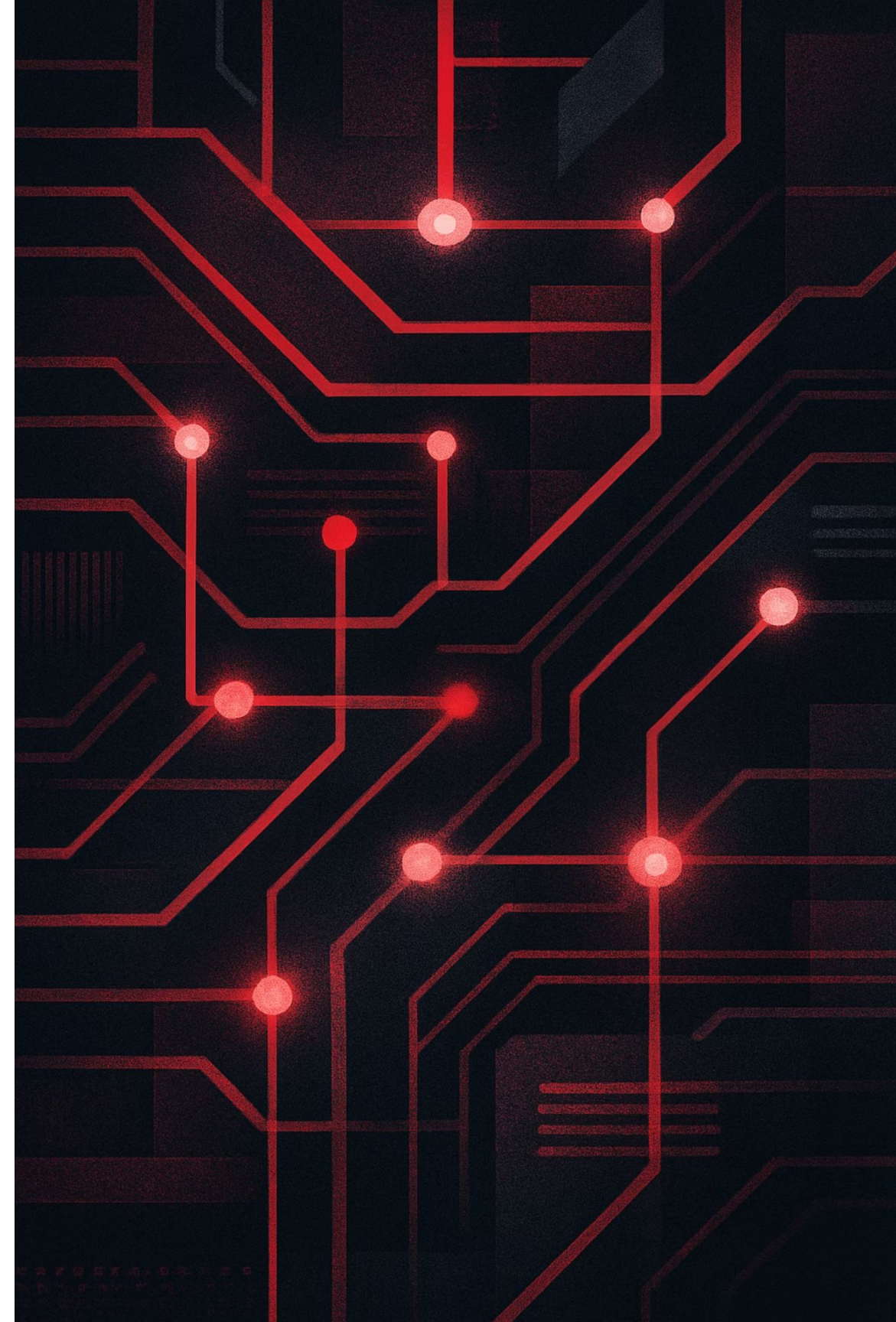
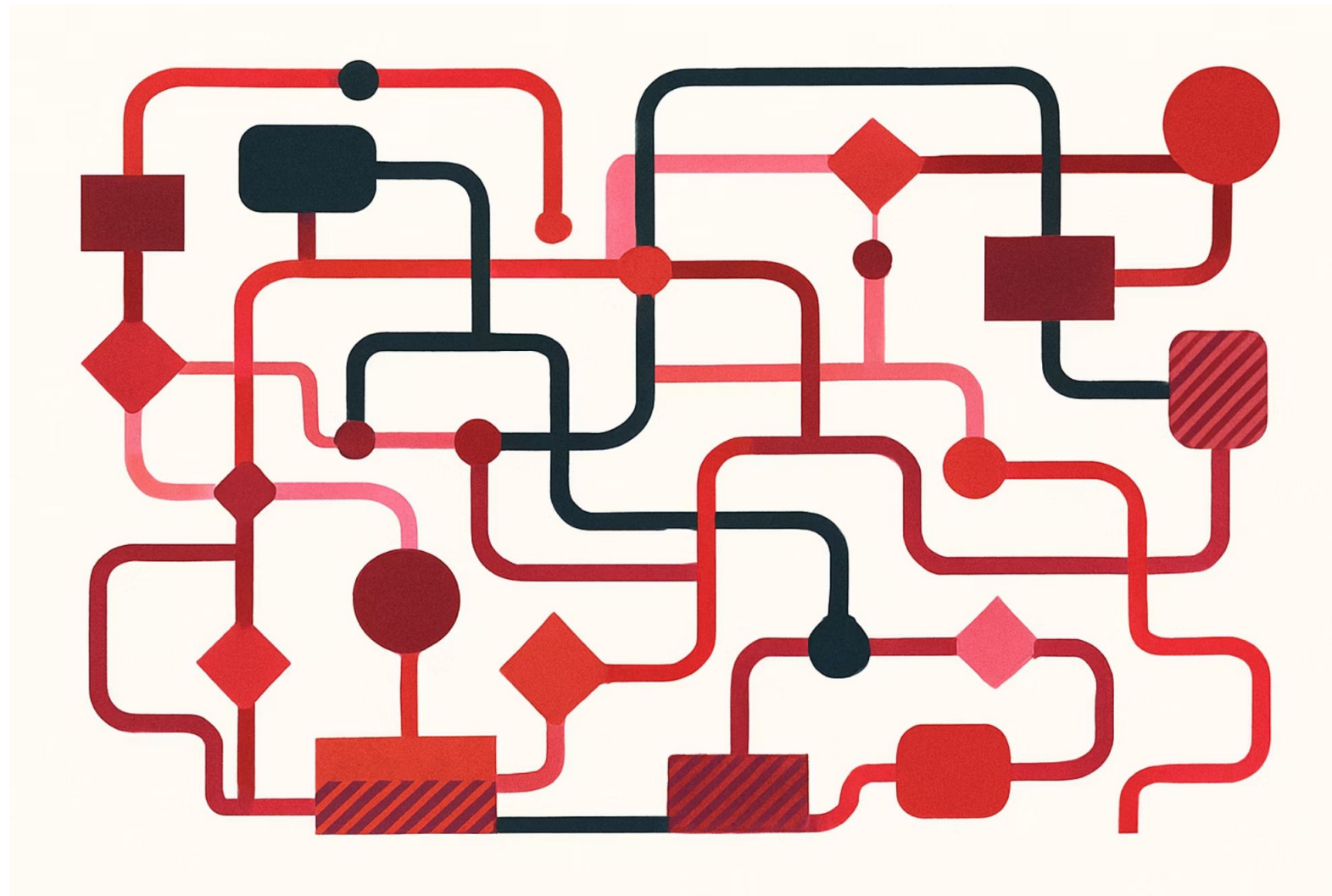


AI for Communication and Marketing

Apr 15, 2026 – Lab lecture #6: Propensity Models | Churn &
Repurchase



What Are Propensity Models?



A **propensity model** is a predictive tool that estimates the probability a customer will take a specific action — using historical behavior to score future likelihood.

Both model types sit at the heart of the **Customer Lifecycle**, shifting marketing from reactive to proactive.

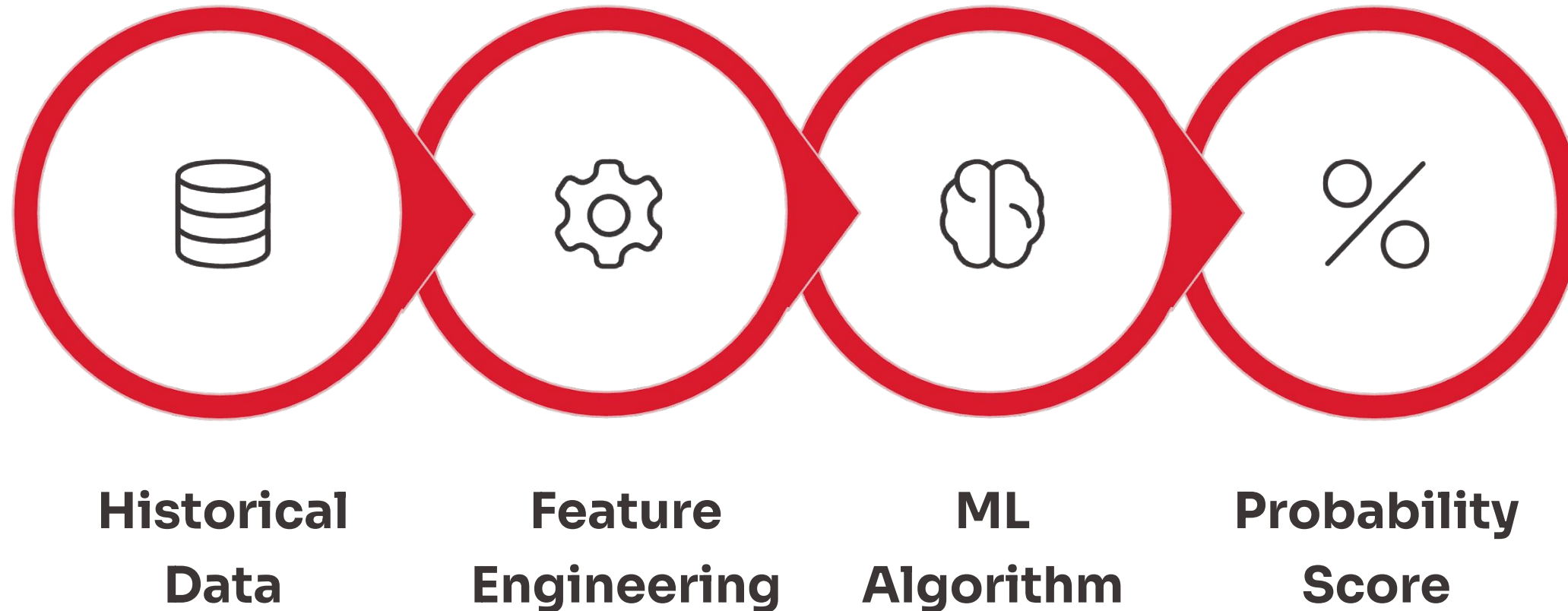
Churn Models

Identify customers at risk of leaving — enabling timely retention interventions.

Repurchase Models

Predict who is likely to buy again — powering loyalty, upsell, and cross-sell campaigns.

From Raw Data to Probability Scores



Machine learning algorithms — **Logistic Regression** for interpretability, **Random Forest** for robustness, or **XGBoost** for peak performance — ingest engineered features and output a score between 0 and 1 for every customer. A score of 0.82 means an 82% predicted probability of churning.

Building the Model: Step by Step

01

Data Collection, Audit and Analysis

Gather **behavioral** (clicks, sessions), **transactional** (purchase history, recency), and **demographic** (age, location) data. Check for data inconsistency and **handle outliers/missing values**. Perform a **high-level EDA**.

02

Feature Engineering

Transform raw fields into predictive signals — e.g., "days since last purchase," "average order value," "support tickets in 90 days."

03

Model Training

Split data into **training** (~80%) and **test** (~20%) sets. The model learns patterns on training data; performance is validated on unseen test data.

04

Scoring

Deploy the trained model to assign a **0–1 probability score** to every customer in your database — updated on a scheduled cadence.



Model Evaluation: What Marketers Must Know



Accuracy:
Misleading with
class imbalance.
Not recommended.



Precision: Minimize
wasted discounts.
Cost efficiency.



Recall: Catch as
many churners as
possible. Revenue
protection.



F1-Score: Balance
between precision
and recall. Overall
performance.

⚠ In churn datasets, 95% of customers may be loyal. A model that predicts "no churn" for everyone achieves 95% accuracy — yet catches zero churners.

Precision answers: "Of the customers we flagged, how many actually churned?" — minimizing wasted retention spend.

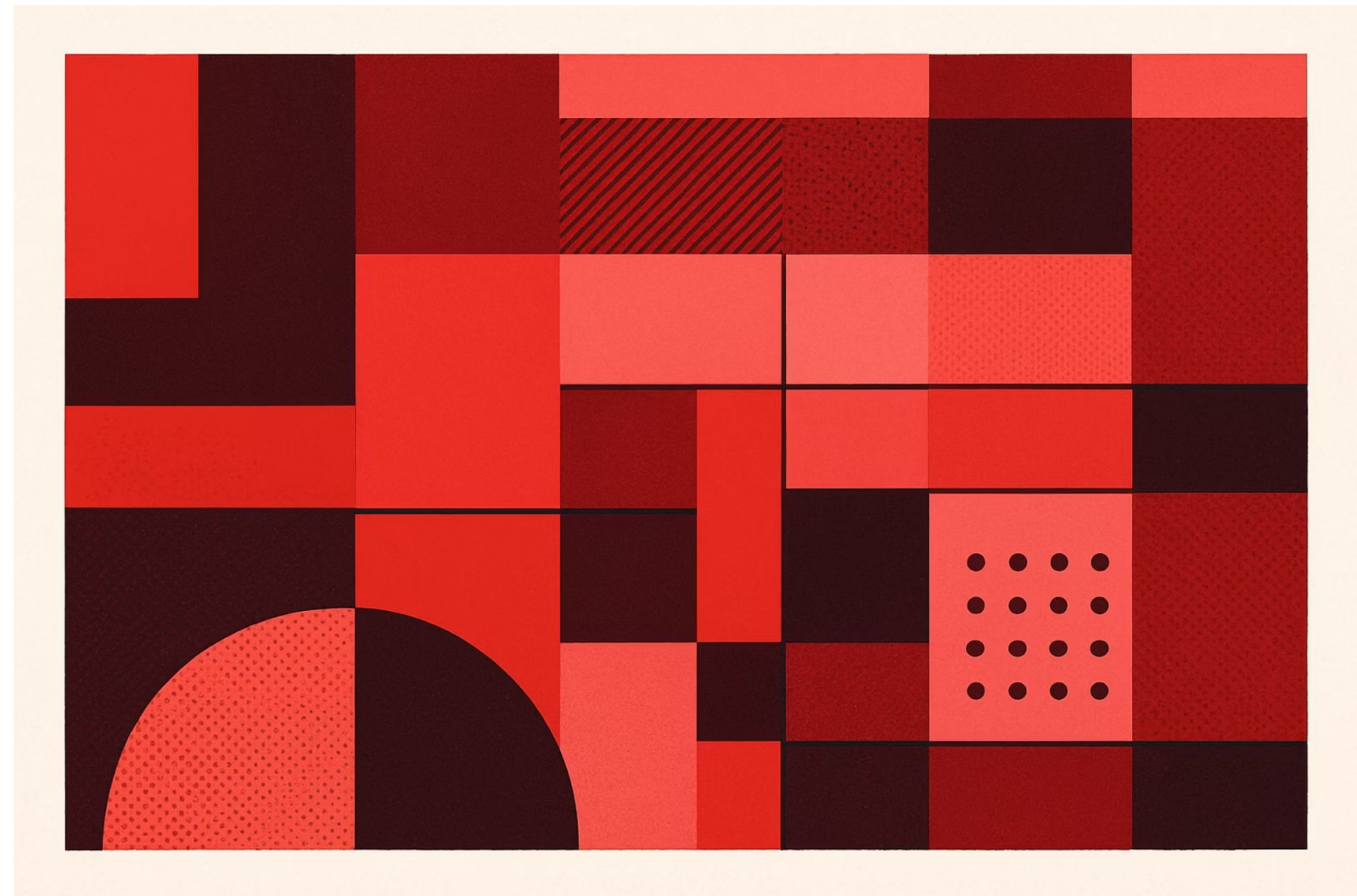
Recall answers: "Of all real churners, how many did we catch?" — maximizing revenue protection.

The **F1-Score** harmonizes both into a single metric, ideal when class imbalance is present.

ROC-AUC & the Confusion Matrix

ROC-AUC Curve

Plots **True Positive Rate** vs. **False Positive Rate** across all thresholds. An AUC of 1.0 = perfect separation; 0.5 = random guessing. Use AUC to compare competing models objectively.



Confusion Matrix

True Positive

Churner correctly flagged → retention offer sent ✓

False Positive

Loyal customer flagged → discount wasted 🚫

False Negative

Churner missed → customer lost ✗

True Negative

Loyal customer ignored → no cost incurred ✓

Hyperparameter Tuning & Model Optimization



The "Knobs & Dials" Concept

Hyperparameters — like tree depth in a Decision Tree or learning rate in XGBoost — are set *before* training. Tuning them shapes how the model learns.



Grid vs. Random Search

Grid Search exhaustively tests every parameter combination. **Random Search** samples combinations randomly — faster and often equally effective for large spaces.

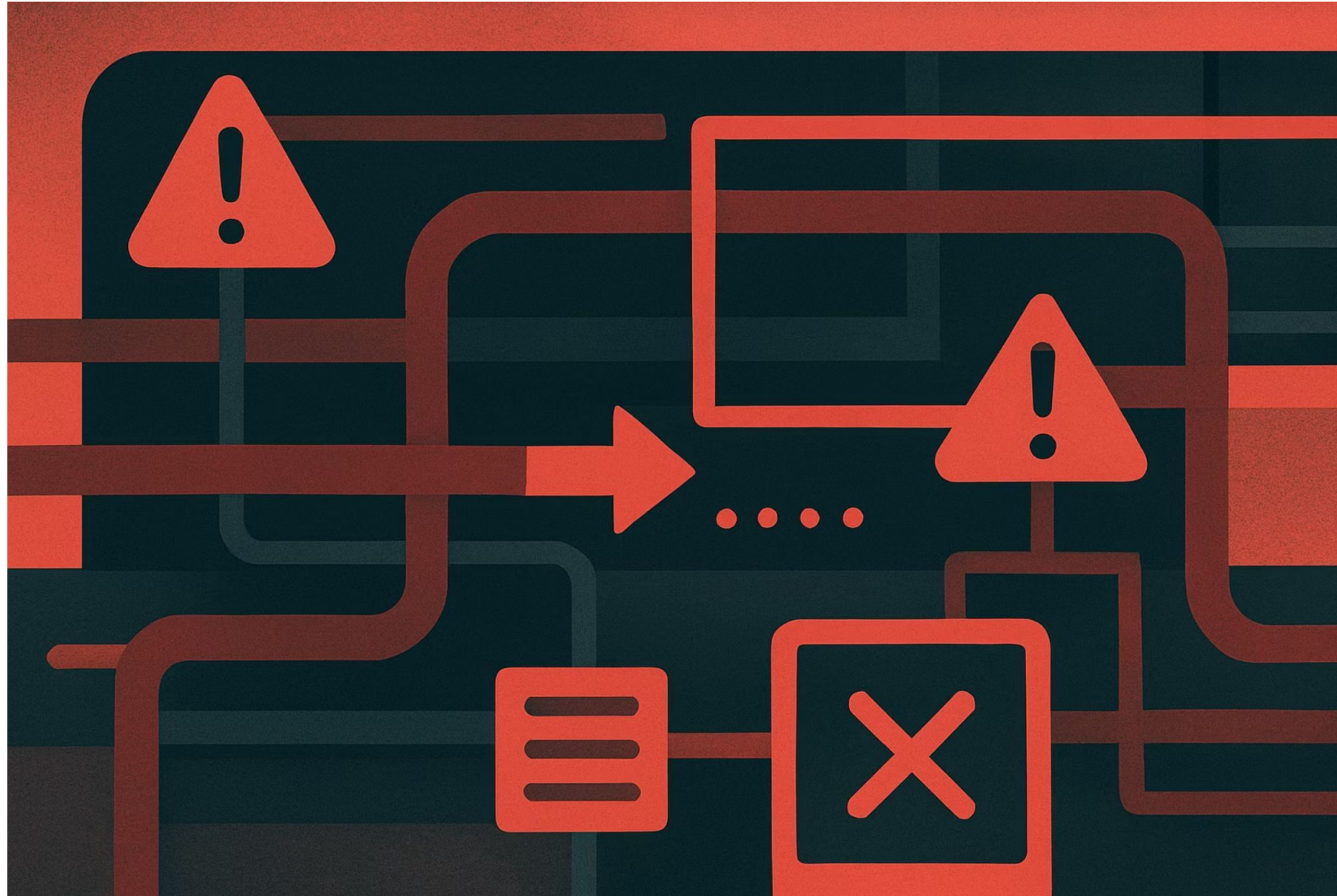


Cross-Validation

Splits training data into k folds, training on $k-1$ and validating on 1. Ensures the model generalizes to new customers and isn't merely memorizing historical patterns (overfitting).



Critical Challenges & Pitfalls



⚠ Data Leakage

Including variables that *reveal* the outcome during training — e.g., "cancellation request date" — produces artificially inflated scores that fail in production.

⚠ Class Imbalance

Churn events are rare. Techniques like **SMOTE** (oversampling) or **class weighting** prevent models from defaulting to always predicting "no churn."

⚠ Correlation ≠ Causation

A customer buying umbrellas may correlate with churn — but sending them a discount won't stop the rain. Validate business logic before acting on model signals.

Business Evaluation & ROI

From Score to Action

Segment customers into **deciles** (top 10%, next 10%...) by propensity score. Concentrate retention spend on the top 2–3 deciles for maximum lift.

Cost-Benefit Framework

 **Lift Charts** measure how much better your model performs versus random targeting — a lift of 3× means your top decile contains 3× more churners than chance.

Decile Analysis Logic

Rank all customers 1–10 by score. Calculate actual churn rate per decile. Plot cumulative gains to visualize where intervention ROI peaks.

A well-tuned model concentrates 60–70% of all churners in the top two deciles — letting you act on 20% of the base to recover the majority of at-risk revenue.

False Negative Cost

Lost customer LTV. The most expensive error.

False Positive Cost

Discount given to a customer who would have stayed anyway — recoverable but erodes margin.

Hands-on Session: Your Churn model challenge

E-Commerce Customer Churn

File: `ecommerce_customer_churn_dataset.csv`

Rows: ~50 000

Columns: 25

The dataset captures behavioural, demographic, and transactional features of e-commerce customers, alongside a binary label indicating whether each customer churned.

Hands-on Session: Your Churn model challenge

01

EDA

Load the data and try to answer four key questions:

1. What does the data look like? (shape, first rows, summary statistics)
2. Which columns have missing values and how many?
3. How is the target variable `Churned` distributed?
4. Are there any obvious outliers or impossible values?

03

Modeling

Train the model/models (please include a XGBoost) and evaluate it using Accuracy, Precision, Recall, F1-score, AUC/ROC score and Lift. Calculate feature importance.

02

Data Cleaning and Pre-processing

- Handle outliers and missing values
- Categorical variables encoding
- Stratified training/test split

04

Hyperparameters tuning

Use Randomized Search/Grid Search to tune the hyperparameters and compare the results with the baseline model's. What model would you use in production?